PRST (09/09/2021)

$P(X=x, Y=y)$ - joint probability mass function

where X, Y are two discrete rvs.

X, Y are two continuous rvs.

$$F_{X,Y}(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(u,v)\,du\,dv$$

joint distribution function

$F(x) = P(X \le x)$

$f(u,v)$ — joint density function.

Marginal distribution

$$G(x) = P(X \le x) = \int_{-\infty}^{x}\left[\int_{-\infty}^{\infty} f(u,v)\,dv\right] du$$

$g(x)$ — pdf of X $\quad \frac{dG(x)}{dx} = g(x)$.

$$Y \to H(y) = P(Y \le y) = \int_{-\infty}^{y}\left[\int_{-\infty}^{\infty} f(u,v)\,du\right] dv$$

distribution function of Y

marginal pdf $h(y) = \frac{dH(y)}{dy}$

~~jt~~ joint density function $f(x,y) = \frac{d^2 F(x,y)}{dx\,dy}$

<u>Ex</u> Suppose the jt pdf of X, Y is

$$f(x,y) = \begin{cases} x^2 + \frac{xy}{3} & 0<x<1, \ 0<y<2 \\ 0 & \text{o.w.} \end{cases}$$

$F\left(\frac{1}{2}, 1\right) = ?$

Sol: $F\left(\frac{1}{2}, 1\right) = P\left(X \le \frac{1}{2}, Y \le 1\right)$

$$= \int_0^{1/2}\int_0^{1}\left(x^2 + \frac{xy}{3}\right) dx\,dy$$

$$= \frac{1}{16}$$

$$F_X\left(\frac{1}{2}\right) = P\left(X \le \frac{1}{2}\right) = \int_0^{1/2}\int_0^{2}\left(x^2 + \frac{xy}{3}\right) dx\,dy = \frac{1}{6}$$

Conditional distribution

$P(X \le x \mid Y = y)$, $P(Y \le y \mid X = x)$

Conditional density function of $X \mid Y = y$

$$f_{X|Y=y}(x) = \frac{f(x,y)}{h(y)}$$

$f(x,y)$ jt pdf, $h(y)$ marginal pdf for Y.

Conditional density function of $Y \mid X = x$

$$f_{Y|X=x}(y) = \frac{f(x,y)}{g(x)}$$

$g(x)$ - marginal density of X.

$$P(X \le x_0 \mid Y = y) = \int_{-\infty}^{x_0} f_{X|Y=y}(x)\,dx$$

$$P(Y \le y_0 \mid X = x) = \int_{-\infty}^{y_0} f_{Y|X=x}(y)\,dy$$

X and Y are independent

$$P(X \in A, Y \in B) = P(X \in A) \times P(Y \in B).$$

$$P(X \in A \mid Y \in B) = P(X \in A).$$

$$f(x,y) = g(x)\,h(y).$$

$$F(x,y) = P(X \le x, Y \le y) = P(X \le x)\,P(Y \le y)$$

$= G(x)\,H(y)$

$= P(X = \ldots)$
$= G_2(x) \times H(y).$

$E(XY) = E(X) \times E(Y).$

$E(g(X) \times h(Y)) = E(g(X)) \times E[h(Y)]$

$Cov(X, Y) = E(XY) - E(X)E(Y)$
$= 0.$

$$Var(X \pm Y) = Var(X) + Var(Y) \pm 2\,Cov(X, Y)$$

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\,Cov(X, Y).$$

$Z = aX + bY$ $\qquad \mu_X = E(X)$, $\mu_Y = E(Y).$

$Var(Z)$

$\mu_Z = E(Z) = a\mu_X + b\mu_Y$

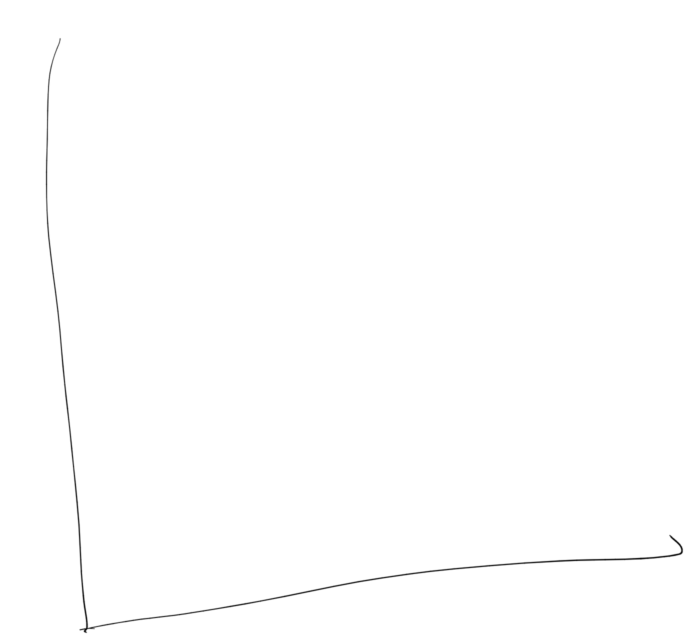
$Var(Z) = E(Z - \mu_Z)^2$
$= E(aX + bY - a\mu_X - b\mu_Y)^2$
$= E[a(X - \mu_X) + b(Y - \mu_Y)]^2$
$= E(a(X - \mu_X)$
$= a^2 \underbrace{E(X - \mu_X)^2}_{} + b^2 E(Y - \mu_Y)^2 + 2ab \underbrace{E(X - \mu_X)(Y - \mu_Y)}_{}$

Eg. If the jt pdf of X and Y is

$$f(x, y) = \begin{cases} k(1 - x - y), & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{o.w.} \end{cases}$$

(i) $k = ?$ $\quad (k = 6)$
(ii) find marginal dist of X and Y.
(iii) Check if X and Y are independent or not.

$0 < x$
$0 < y$
$x + y < 1$
$y < 1 - x$
$0 < y < 1 - x,$
$0 < x < 1$



~~$g(x)$~~

(ii) $$g(x) = \int_0^{1-x} f(x, y)\, dy$$
$$= \int_0^{1-x} k(1 - x - y)\, dy$$
$$= \frac{k}{2}(1 - x)^2 \quad \text{if } 0 < x < 1$$

$x > 0$

$\boxed{x < 1-y}$

$$= 2$$

$$h(y) = \int_0^{1-y} k(1-x-y)\,dx = \frac{k}{2}(1-y)^2 \quad \text{if } 0 < y < 1$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

$$\Rightarrow \int_0^1 \int_0^{1-x} k(1-x-y)\,\cancel{dx}\,dy\,dx = 1$$

or $$\int_0^1 \int_0^{1-y} k(1-x-y)\,dx\,dy = 1$$

$$f(x,y) \neq g(x)\,h(y)$$

$\therefore$ X ~~are~~, Y are not independent.